Financial Portfolio Optimization by LLM

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Executive Summary Report

Project Goals

1. Can an LLM be used for professional financial optimization given set constraints and guardrails?

2. Can Claude AI generate Python notebooks, documents and slide decks to be used in industry?

Project Method

1. Step 1: Create a hand-coded Python notebook that solves for optimal weights using Sequential Least Squares Programming (SLSQP) with given expected returns
2. Step 2: Iteratively prompt Claude Code to execute Step 1 and cross-review generated code against hand-code
3. Step 3: Iteratively prompt Claude AI to generate slide deck and executive summary for python notebook(s)
4. Step 4: Review and clean up Slide Deck and Executive Summary Document

Python Code Methodology

μ Expected Returns

Analyst-supplied forward-looking views (NOT historical means)

Σ Calculate Variance and Covariance

Historical daily log returns (2021–present)

➤ Maximize Sharpe Ratio

Maximize Sharpe Ratio via SLSQP solver fully invested, 4%–24% per ETF

User-defined Guardrails

- Minimum of 4% investment to maintain portfolio diversification
- Maximum of 24% investment to minimize portfolio over-concentration
- 7 ETFs selected can be changed

Risk-Free Rate: 5.00% | Benchmark: Equal-Weight (1/N)

Final Python Notebook Generation Prompt

“Generate a notebook named ETF_Target_Weight_Optimization. Solve for optimal weights of a portfolio of a certain number of assets. The weights are bound between 4% and 24% of the portfolio. All weights must add up to 100%. The Assets in question include IWV, VCIT, SPTL, VEA, IEMG, TIP, and JNK. Pull historical Close prices from 2021-01-01 to today for financial data. Calculate the portfolio sharpe’s ratio and provide optimal weights based on a maximum sharpe’s ratio. Compare against a portfolio of equal weighted assets. Use these expected returns to calculate portfolio weights per each etf with a given risk free rate of 0.5
US Equity: 8% |Investment Grade Bonds: 4.9% | Long Term Government Bonds 5.1% | Developed Non-US Equity: 7.9% | Emerging Market Equity: 8% | TIPS: 4.7% | High Yield Bonds: 6.6% “

Portfolio Metrics

Ticker	Asset Class	Expected Return (μ)	Hist. Volatility	Sharpe Ratio
IWV	US Equity (Russell 3000)	**8%**	~14%	**0.21**
IEMG	Emerging Market Equity	**8%**	~17%	**0.18**
VEA	Developed Non-US Equity	**7.9%**	~14%	**0.21**
JNK	High-Yield Bond	**6.6%**	~9%	**0.18**
SPTL	Long-Term Government Bond	**5.1%**	~15%	**0.01**
VCIT	Investment-Grade Corp Bond	**4.9%**	~7%	**-0.01**
TIP	Inflation-Protected Bond	**4.7%**	~8%	**-0.04**

Rf = 5.00% | Sharpe = (μ − Rf) / σ
Ticker, Asset Class, and Expected Returns are all user defined metrics
Historical Volatility and Sharpe Ratio are both calculated metrics

Portfolio Performance Results

Equal-Weight Baseline

6.64%
Expected Return

9.32%
Volatility

0.178
Sharpe Ratio

Max-Sharpe Optimized

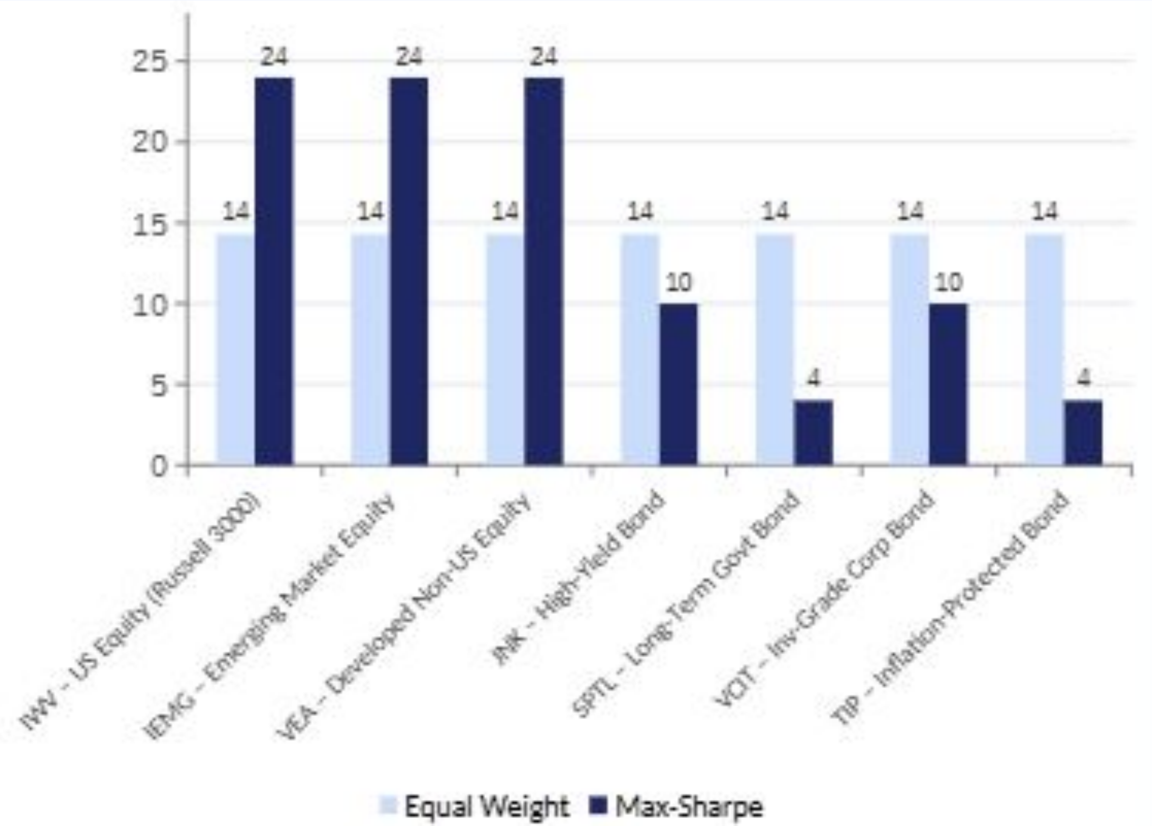
7.11% +0.47%
Expected Return

8.82% −0.50%
Volatility

0.238 +0.060
Sharpe Ratio

33.7% Sharpe Improvement vs Equal-Weight Baseline

Optimal Portfolio Weights



▲ 24%

IWV / IEMG / VEA

▼ 4%

SPTL / TIP

○ 10%

JNK / VCIT

Bounds: Min 4% | Max 24% per ETF

Final Executive Report and Slide Deck Prompt:

“Generate a google slide deck and a google doc that summarizes the main points from the notebook provided. Ensure these decks and documents are professional and really emphasize the results of the portfolio metrics. Make sure the font is well sized and anything wordy should not be and instead should be short, concise, and to the point. don't be repetitive and redundant with slides either. “

Project Goals Completed

1. Can an LLM be used for professional financial optimization given set constraints and guardrails?
   a. Yes, Claude Code can perform financial optimization tasks such as finding optimal portfolio weights from a given expected return ***after iteratively finding the right prompt***

2. Can Claude AI generate Python notebooks, documents and slide decks to be used in industry?
   a. Yes, Claude AI generates Python notebooks, executive summaries, and slide decks, but these materials would ***require a final human-led review before official use***

Additional Insights

1. Can an LLM be used for professional financial optimization given set constraints and guardrails?
   a. Yes, Claude Code can perform financial optimization tasks such as finding optimal portfolio weights from a given expected return ***after iteratively finding the right prompt***

2. Can Claude AI generate Python notebooks, documents and slide decks to be used in industry?
   a. Yes, Claude AI generates Python notebooks, executive summaries, and slide decks, but these materials would ***require a final human-led review before official use***